

Jiaqi Ge

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EDUCATION

Guangdong Technion - Israel Institute of Technology (English-taught Program)

Sep 2023 - Present

B.Sc. in Mathematics with Computer Science (GPA: 89.7/100)

PUBLICATIONS

Manuscript on repository knowledge acquisition for software issue resolution

Co-author, under submission

WORK EXPERIENCE

Undergraduate Teaching Assistant - Linear Algebra & Calculus

Mar 2026 - Present

Guangdong Technion - Israel Institute of Technology, Shantou, China

- Facilitate student learning in weekly workshops by providing guidance on complex mathematical problems and theorems.
- Assist students in mastering foundational concepts of Linear Algebra and Calculus through interactive problem-solving sessions.

Undergraduate Teaching Assistant - Python

Oct 2025 - Jan 2026

Guangdong Technion - Israel Institute of Technology, Shantou, China

- Assisted in teaching the introductory Python course and supervised lab sessions.
- Supported students with debugging code and understanding core programming concepts.

RESEARCH EXPERIENCE

Research Assistant - Repository QA for Software Issue Resolution

Feb 2026 - Present

Shanghai Jiao Tong University, advised by Xiaodong Gu

- Contributed to manuscript writing and revision, including related work, experimental results, method analysis, ablation discussion, and figure/result interpretation.
- Designed and executed baseline vs. QA-augmented experiments for repository-level software issue resolution.
- Built a judge-based evaluation pipeline for question-level, set-level, and pairwise assessment, and verified the scoring and aggregation process.
- Conducted fail-to-pass trajectory analysis, including repair-stage segmentation, QA-usage analysis, and patch-alignment analysis, and prepared supporting figures and reports.

Research Assistant - Mixture-of-Experts (MoE)

Apr 10, 2026 - Present

Abu Dhabi AI, advised by Prof. Lijie Hu

- Working on a research project related to Mixture-of-Experts (MoE).

Research Assistant - Explaining AI Models through Metamorphic Property Generation

Oct 2025 - Feb 2026

Guangdong Technion - Israel Institute of Technology, advised by Prof. Nazareno Aquirre

- Built a grammar-based fuzzing pipeline to generate multi-step text perturbation chains (typos/punctuation/entity swaps/negation.....) using Checklist.
- Implemented candidate metamorphic-property generation and checking (INV invariance / DIR directional) to evaluate model behavior under k-step transformations.
- Integrated a HuggingFace sentiment-analysis model with Checklist to run automated INV/DIR test suites at scale; summarized experimental findings for research documentation.

Research Assistant - Numerical Methods for Boundary Value Problems (Project Score: 95/100) Aug 2025 - Nov 2025

Advisor: Prof. Antti Rasila

- Conducted research on numerical solutions for boundary value problems (BVPs) in MATLAB.
- Reviewed classical theory including existence/uniqueness, Green's functions, and Sturm-Liouville problems.
- Compared shooting, finite-difference, and spectral methods for linear and nonlinear BVPs.
- Used MATLAB Chebfun toolbox to solve nonlinear systems, singular Sturm-Liouville problems, and chaotic dynamical systems.
- Authored a 20-page report titled "Solving Boundary Value Problems Using Chebfun".

Research Assistant - Integration of Multi-Age Clocks for Alzheimer's Disease Diagnosis May 2024 - Jun 2025

Advisor: Dr. Shibiao Wan | University of Nebraska Medical Center (UNMC)

- Integrated brain-aging and epigenetic aging models to support Alzheimer's disease (AD) diagnosis, and analyzed relationships among different age clocks for age-related disease prediction.
- Contributed to the implementation and code review of a multi-modal age-clock integration framework combining DNA methylation and neuroimaging data.
- Conducted literature reviews on biological aging biomarkers and age estimation models.

SKILLS

Programming: Python, C, C++, Java, MATLAB, Assembly

Languages: Chinese (Native), English (Fluent)

AWARDS

Scholarship - 28,500 USD (2023)

Awarded by Guangdong Technion - Israel Institute of Technology for performance in the Chinese National College Entrance Examination (Gaokao).